

MAKE A



DRAWING TOOL



p5.js

```

function setup() {
  createCanvas(400, 400);
  background(220);
}

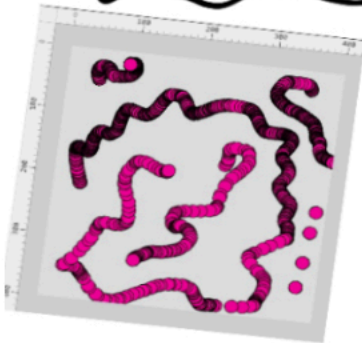
function draw() {
  if (mouseIsPressed) {
    fill(255, 0, 150);
    ellipse(mouseX, mouseY, 20, 20);
  }
}

```

createCanvas (400, 400)
 ↑ ↑
 width height

DID YOU KNOW?

You can use windowWidth, windowHeight to make the canvas auto-adjust



Your turn: Change
 some numbers. Press
 ▶ to run the code
 again. What do you
 notice?

COLORS

These functions take a color:

`background()` ← paints whole canvas

`fill()` ← paints inside shapes

`stroke()` ← paints the outline

GRAYSCALE: one number

`fill(0)` ← black

`fill(128)` ← gray

`fill(255)` ← white

RGB : three numbers

`fill(255, 0, 0)` ← red

`fill(0, 255, 0)` ← green

`fill(255, 200, 0)` ← yellow-orange

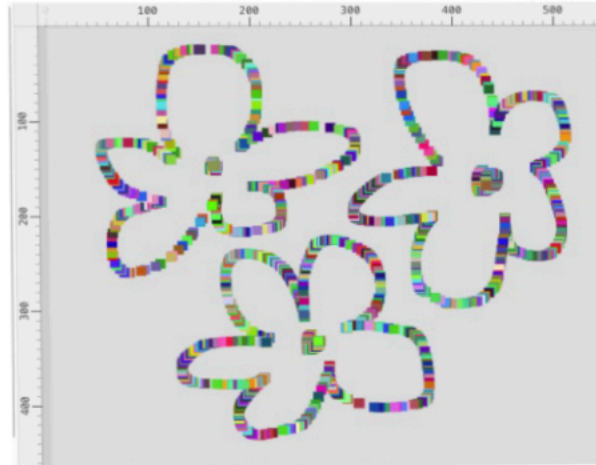
↑ ↑ ↑
red green blue
amount amount amount

NOTE: Numbers range from 0-255. You're mixing light, not paint, so `255, 255, 255` gives **white**.

See-through Colors (alpha)

`fill(0, 100)` ← see-through black

`fill(255, 0, 0, 50)` ← ghostly red



```
function setup() {  
  createCanvas(windowWidth, windowHeight);  
  background(220);  
}  
  
function draw() {  
  if (mouseIsPressed) {  
    noStroke();  
    fill(random(255), random(255), random(255));  
    rect(mouseX, mouseY, 10, 10);  
  }  
}
```

LOOKS LIKE CONFETTI!

MEET random()

Each time it runs, `random(255)` gives back a surprise number. This ↑ one gives back a number from 0 up to ~255.

Pick a range : `fill(random(200, 255), 0, random(100));`

min max implied zero min

Red value randomly chosen between 200-255 green Value always 0 blue value randomly chosen between 0-100

SHAPES

`rect(mouseX, mouseY, 10, 20)`

These can take static x- or y-coordinates or use `mouseX` & `mouseY` to draw the Shape where the cursor is.

width of the rectangle height of the rectangle

`rect(x, y, w, h)` ← in other words ☺

`ellipse(x, y, w, h)`

`line(x1, y1, x2, y2)`

`triangle(x1, y1, x2, y2, x3, y3)`

`quad(x1, y1, x2, y2, x3, y3, x4, y4)`

You can use `random()` for any value



SWITCH BETWEEN BRUSHES (

- ① Make your brush into a function :

```
function confettiBrush () {  
  if (mouseIsPressed) {  
    noStroke();  
    fill(random(255), random(255), random(255));  
    rect(mouseX, mouseY, 10, 10);  
  }  
}
```

- ② Great! Now, "Call" the function in draw()

```
function draw () {  
  confettiBrush();  
}
```

- ③ make
at least two
different brushes
into functions.

```
1 function setup() {  
2   createCanvas(windowWidth, windowHeight);  
3   background(220);  
4 }  
5  
6 function draw(){  
7   confettiBrush();  
8 }  
9  
10 function confettiBrush(){  
11  
12   if(mouseIsPressed){  
13     noStroke();  
14     fill(random(255), random(255), random(255));  
15     rect(mouseX, mouseY, 10, 10);  
16   }  
17 }  
18  
19
```

④ KeyPressed()

```
function KeyPressed() {  
  if (key == "1") {  
    currentBrush = 1;  
  }  
  if (key == "2") {  
    currentBrush = 2;  
  }  
}
```

Add
more as needed

KeyPressed runs
when you press a key.

key holds which key you pressed. "1"
goes in quotes because a key is a character,
not a number.

⑤ Then use draw() to check which brush was chosen.

```
function draw() {  
  if (currentBrush == 1) {  
    ConfettiBrush();  
  }  
  if (currentBrush == 2) {  
    bubbleBrush();  
  }  
}
```

Add more
as
needed

```

1  let currentBrush = 1;
2
3  function setup() {
4    createCanvas(400, 400);
5    background(220);
6  }
7
8  function draw(){
9
10     if(currentBrush == 1){
11       bubbleBrush();
12     }
13
14     if(currentBrush == 2){
15       confettiBrush();
16     }
17   }
18
19   function bubbleBrush(){
20
21     if(mouseIsPressed){
22       noStroke();
23       fill(120,180,255,80);
24       ellipse(mouseX,mouseY,25,25);
25     }
26   }
27
28   function confettiBrush(){
29
30     if(mouseIsPressed){
31       noStroke();
32       fill(random(255), random(255), random(255));
33       rect(mouseX,mouseY,10,10);
34     }
35   }
36

```

Don't forget to add this variable to keep track of which key was pressed.



This zine goes with this tutorial!

Scan here

openprocessing.org/amy-bw/2891452